

Huang Kuan Po, Ph.D.

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Research Summary

- My research centers on **audio generation**, building models that synthesize high-quality audio efficiently. I also work on **text-to-speech (TTS)** and **speech understanding**, spanning generative modeling and speech representation learning.

Employment History

- Jun 2026 – Oct 2026 **Research Scientist Intern, Meta**
 - Inference-time emotional speech generation.
- Dec 2025 – Apr 2026 **Research Scientist Intern, Bytedance Seed-Speech Multimodal Interaction, San Jose, California.**
 - Executed post-training pipelines for Large Audio Language Models (LALMs) to advance capabilities in speech and audio understanding.
 - Conducted large-scale distributed training across a high-performance cluster of 128+ H-series GPUs.
- Jul 2024 – Oct 2024 **Applied Scientist Intern, Amazon AGI Speech & Audio, Bellevue, Washington.**
 - Delivered a state-of-the-art diffusion-based model that excels in both performance and inference speed for text-to-audio generation. (Paper accepted by ICML 2025).
 - 🔗 Project page
 - Delivered a decoder-only audio LM for text-to-audio generation. (Paper accepted by ICML 2025).
- Apr 2025 – Jun 2025 **Applied Scientist Intern, Amazon AGI Speech & Audio, Bellevue, Washington.**
 - Improved inference speed of 8.5x for text-to-audio generation models while preserving performance with energy-distance and representation distillation. (Submit to NeurIPS 2026).
- Sep 2022 – Nov 2025 **Ph.D. Student, AICS (ASUS) Ph.D. Program, Taipei, Taiwan.**
 - Domain adaptive methods for SSL speech models. (Paper accepted by Interspeech 2022).
 - 🔗 Model
 - Distilling SSL speech models into smaller versions. (Paper accepted by SLT 2022 and ICASSP 2023).
 - 🔗 Code
 - Code-switching evaluation for SSL speech models. Proposed new code-switch speech datasets. (Paper accepted by ICASSP 2024).
 - 🔗 Code & dataset

Education

- Sep 2022 – Jun 2026 **Ph.D. in Computer Science and Information Engineering (Direct Admission)**
National Taiwan University, Taipei, Taiwan
Advisors: Prof. Lin-shan Lee, Prof. Hung-yi Lee
- Sep 2020 – Jun 2022 **M.S. in Computer Science and Information Engineering (Transfer to Ph.D.)**
National Taiwan University, Taipei, Taiwan

Education (continued)

Sep 2016 – Jun 2020

■ **B.S. in Computer Science and Information Engineering**
National Taiwan University, Taipei, Taiwan

Research Publications

- 1 H.-L. Chung, **K.-P. Huang**, B.-R. Lu, and H.-Y. Lee, “Fréchet distance loss on speech representations for text-to-speech synthesis,” 2026.
- 2 **K.-P. Huang**, H.-L. Chung, B.-R. Lu, and H.-Y. Lee, “Fdaudio: Meanflow-anchored fréchet-distance post-training for one-step text-to-audio generation,” 2026.
- 3 **K.-P. Huang**, B.-R. Lu, B. Kim, L. Mihee, F. Zalan, K. Renard, Q. Tang, S. G. Ver, H.-y. Lee, C.-C. Kao, and C. Wang, “Fast text-to-audio generation with one-step sampling via energy-scoring and auxiliary contextual representation distillation,” *arXiv preprint arXiv:2605.00329*, 2026.
- 4 K.-H. Lu, Z. Chen, S.-W. Fu, C.-H. H. Yang, S.-F. Huang, C.-K. Yang, C.-E. Yu, C.-W. Chen, W.-C. Chen, C.-y. Huang, and **K.-P. Huang**, “Destaz. 5-audio: Toward general-purpose large audio language model with self-generated cross-modal alignment,” *IEEE Transactions on Audio, Speech and Language Processing*, 2026.
- 5 **K.-P. Huang**, S.-W. Yang, H. Phan, B.-R. Lu, B. Kim, S. Macha, Q. Tang, S. Ghosh, H.-Y. Lee, C.-C. Kao, et al., “Impact: Iterative mask-based parallel decoding for text-to-audio generation with diffusion modeling,” in *International Conference on Machine Learning*, PMLR, 2025, pp. 26 002–26 019.
- 6 S.-W. Yang, **K.-P. Huang**, B. Kim, Q. Tang, H. Phan, B.-R. Lu, H. Sundar, S. Ghosh, H.-Y. Lee, C.-C. Kao, et al., “Generative audio language modeling with continuous-valued tokens and masked next-token prediction,” in *International Conference on Machine Learning*, PMLR, 2025, pp. 70 793–70 812.
- 7 **K.-P. Huang**, C.-K. Yang, Y.-K. Fu, E. Dunbar, and H.-y. Lee, “Zero resource code-switched speech benchmark using speech utterance pairs for multiple spoken languages,” in *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, 2024, pp. 10 006–10 010.
- 8 F. Ritter-Gutierrez, **K.-P. Huang**, J. H. Wong, D. Ng, H.-y. Lee, N. F. Chen, and E.-S. Chng, “Dataset-distillation generative model for speech emotion recognition,” in *Proc. Interspeech 2024*, 2024, pp. 2640–2644.
- 9 C.-K. Yang, **K.-P. Huang**, and H.-y. Lee, “Do prompts really prompt? exploring the prompt understanding capability of whisper,” in *2024 IEEE Spoken Language Technology Workshop (SLT)*, IEEE, 2024, pp. 1–8.
- 10 **K.-P. Huang**, T.-H. Feng, Y.-K. Fu, T.-Y. Hsu, P.-C. Yen, W.-C. Tseng, K.-W. Chang, and H.-Y. Lee, “Ensemble knowledge distillation of self-supervised speech models,” in *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, 2023, pp. 1–5.
- 11 **K.-P. Huang**, Y.-K. Fu, Y. Zhang, and H.-y. Lee, “Improving distortion robustness of self-supervised speech processing tasks with domain adaptation,” in *Proc. Interspeech 2022*, 2022, pp. 2193–2197.
- 12 **K.-P. Huang**, Y.-K. Wu, and H.-y. Lee, “Multi-accent speech separation with one shot learning,” in *Proceedings of the 1st Workshop on Meta Learning and Its Applications to Natural Language Processing*, 2021, pp. 59–66.
- 13 **K.-P. Huang**, Y.-K. Wu, Y. Tsao, and H.-y. Lee, “One shot learning for speech separation,” in *ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, 2021, pp. 5769–5773.

Miscellaneous Experience

Awards and Achievements

- 2019  **Yahoo Best Internship Award**
- 2021  **Forward-looking Speech Technology Scholarship**
- 2026  **Phi Tau Phi Award (Rank 1 in Ph.D. at NTU CSIE)**

Peer Reviews

-  **Conferences:** ICML (2026), ARR (2026), NeurIPS (2024, 2025), ICASSP (2022–2025), Interspeech (2021, 2022, 2026), ASRU (2021), ACL (2021), ISCSLP (2022, 2024), ACML (2025).
-  **Journal:** Signal Processing Letters (2025, 2026)